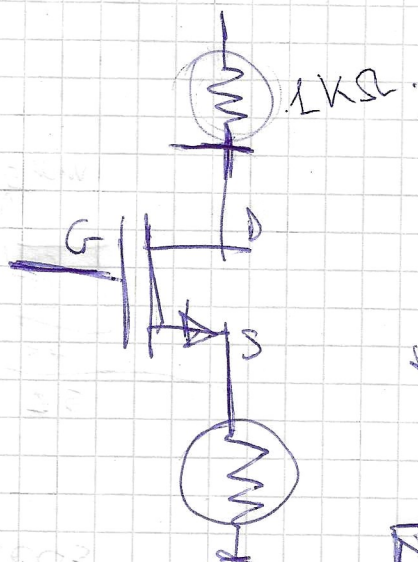
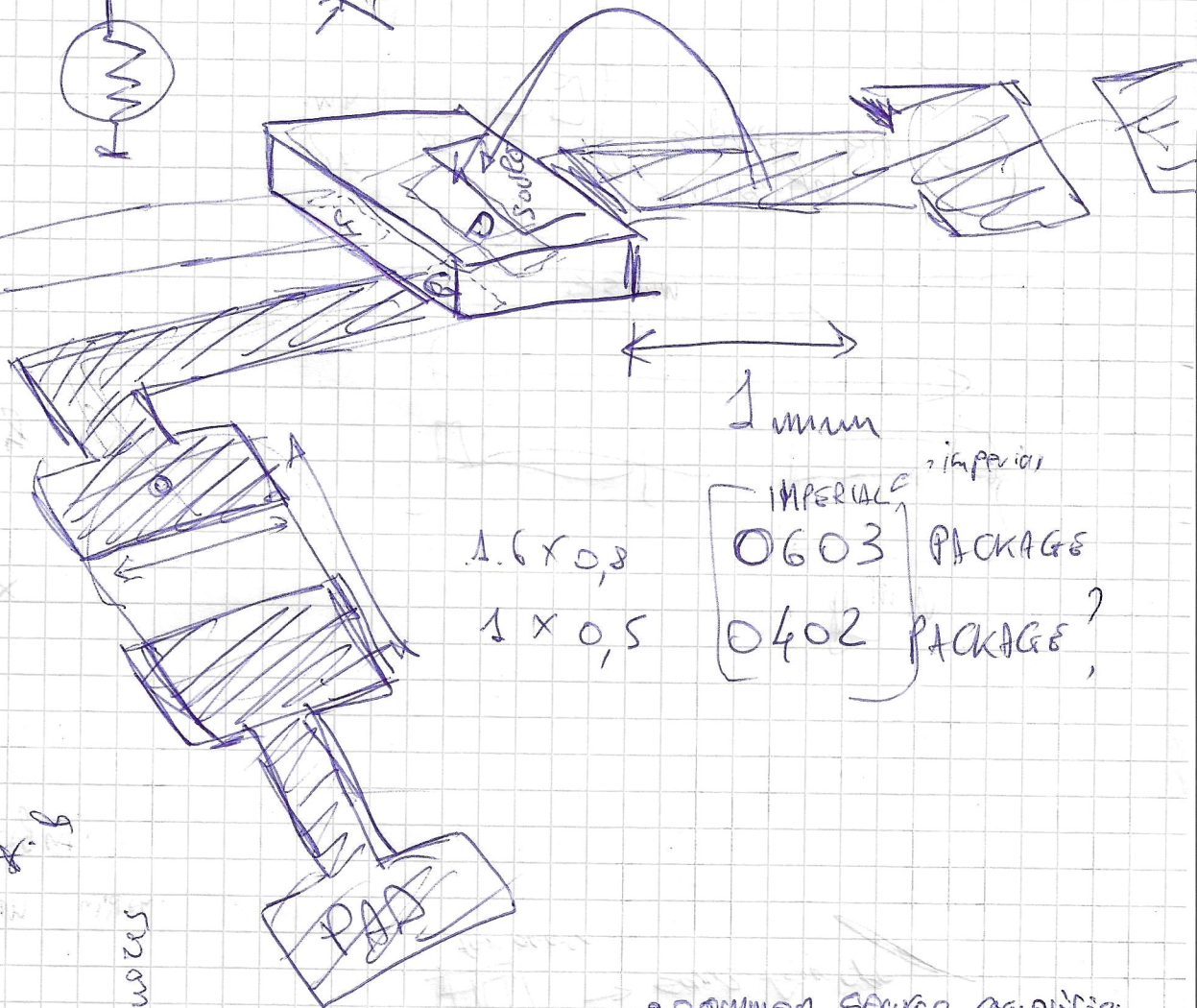
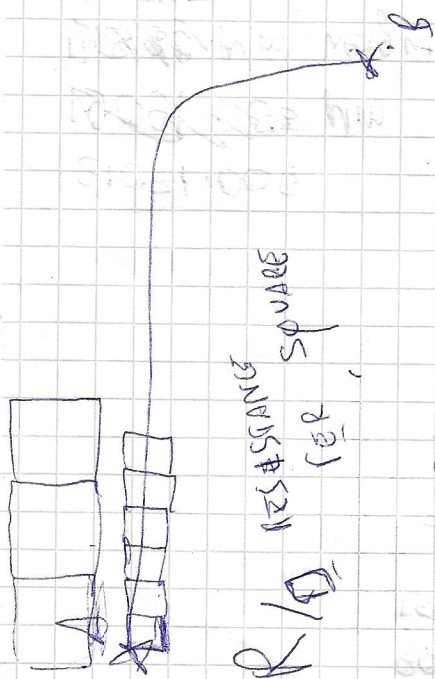




Width $> 10 \mu\text{m}$



IMPERIAL
0603 PACKAGE
0402 PACKAGE?



$R_{total} = R \cdot \# \text{ Squares}$
182/1

- common source amplifier
- common drain amplifier
- characterization of transistor.

$$\left. \begin{array}{l} 60 \mu m \\ 90 \mu m \\ 153 \end{array} \right\} 73 \rightarrow 36.5$$

$$218 - 90 = 138 \rightarrow 69.5$$

$$69 - 60 = 9 \mu m \rightarrow 4.5$$

$$\left(\begin{array}{c} 17.5 \\ 5 \end{array} \right)$$

$$21071.005$$

$$10091.000 \mu m \text{ width}$$

$$14300 \mu m \text{ height}$$

$$14545$$

$$\begin{array}{r} 5000 \times 8000 \\ + 271.03 \\ = 10771.03 \times \end{array}$$

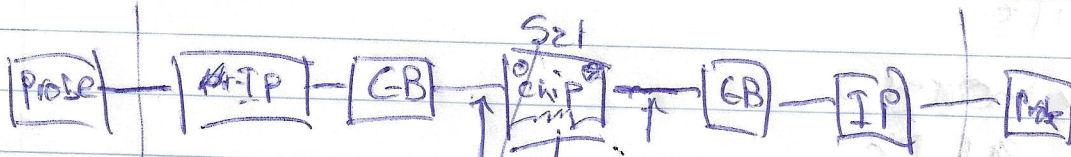
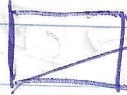
$$\begin{array}{r} 7700 - 4100 \\ = 2100 \end{array} y$$

$$40 \times 10771.003$$

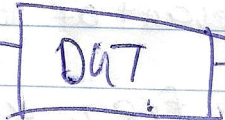


→ same for 90





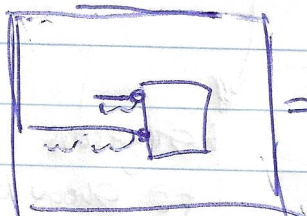
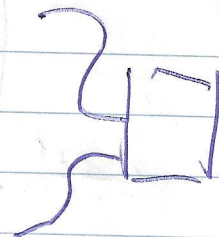
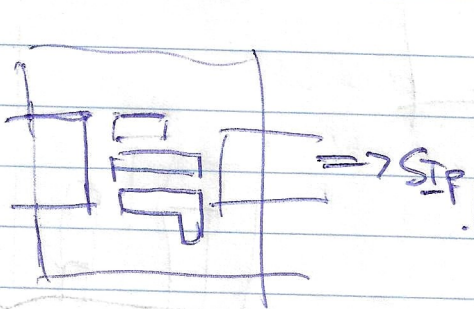
calibration



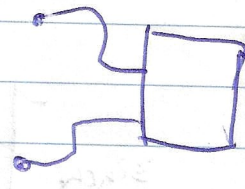
calibration

$$S_{out} = \cancel{S_{IP}} (S_{IP} S_{CB})^2 S_{chip}$$

measure directly



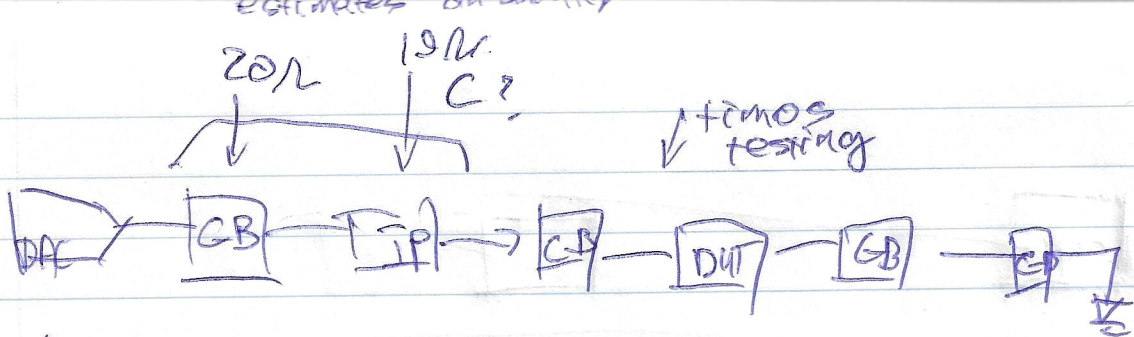
$$\begin{matrix} S_{IP} S_{CB} S_{chip} \\ S_{IP}^2 S_{CB} S_{chip} \end{matrix}$$



have a microstrip version



- known load
- short, open impedance



1MHz

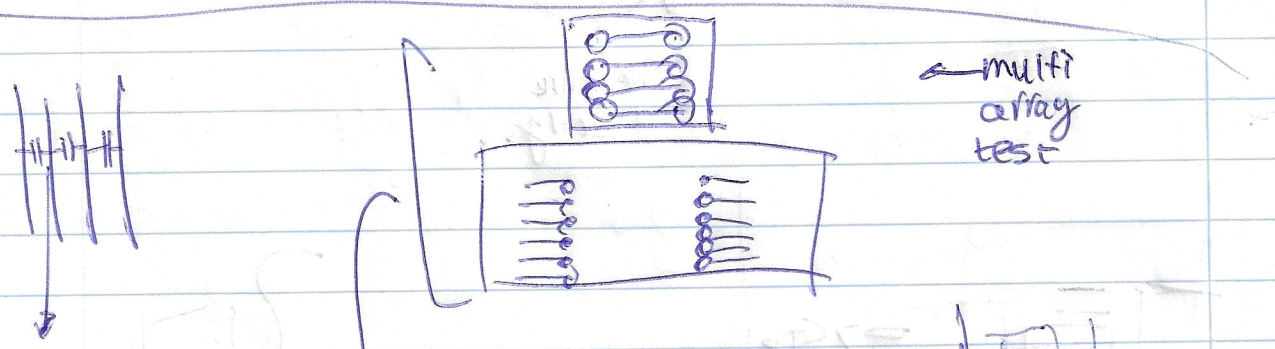
↳ looked at DC

↳ The architecture characterization of coplanar

↳ not relevant at low freq.

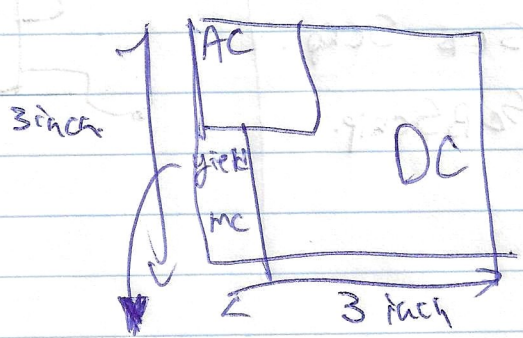
at high rf freq would matter

Sel

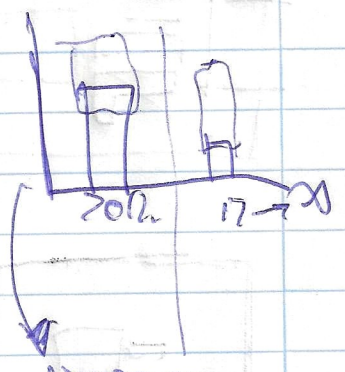


← multi array test

then flip chip & categorize.



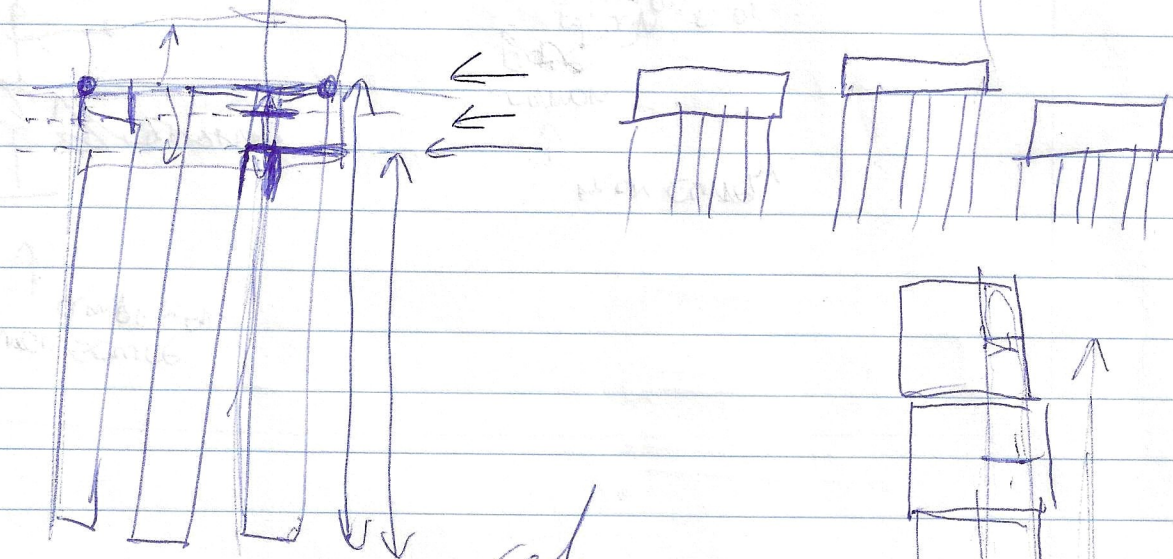
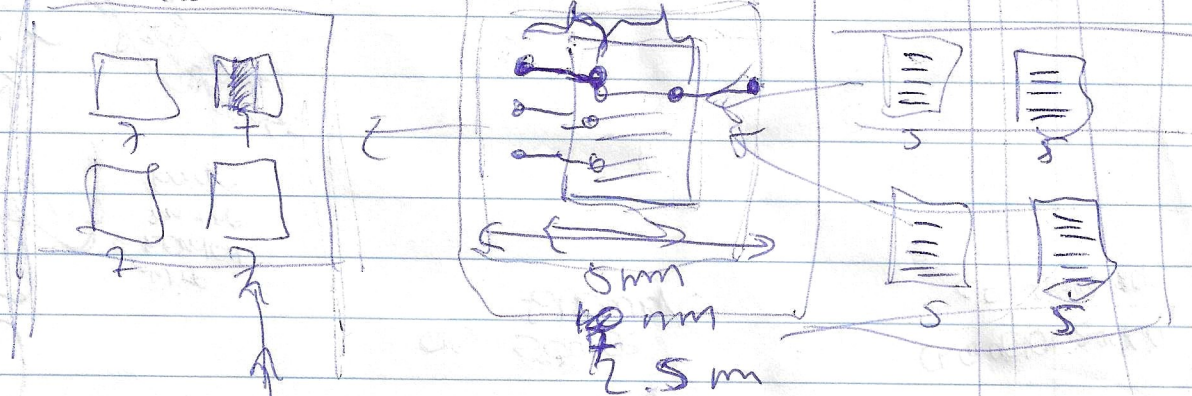
4 interposer mask.



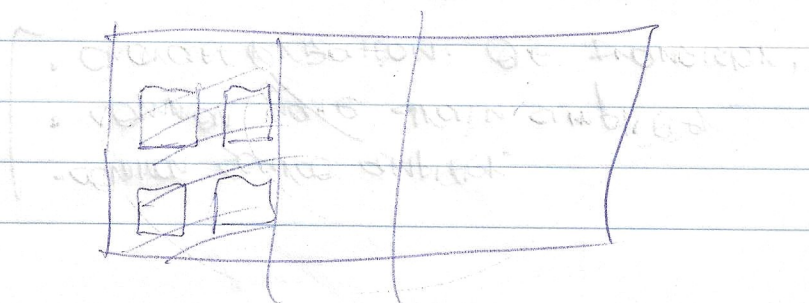
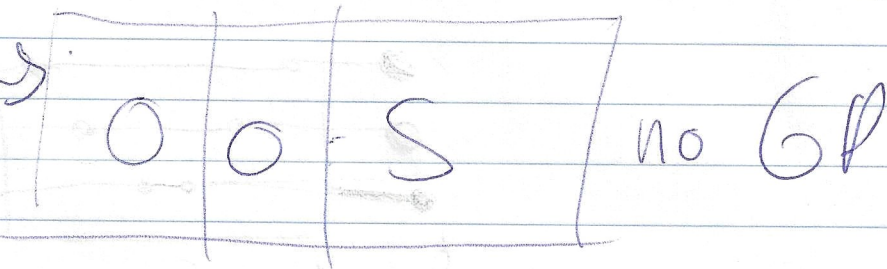
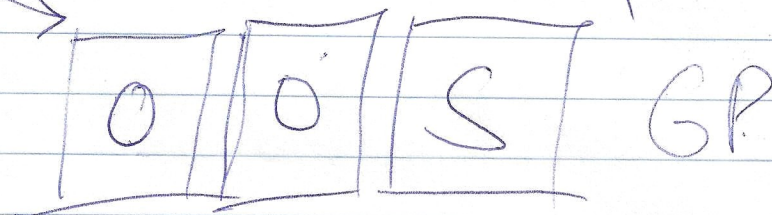
histogram to show cap yield with different heat profiles for bumps.

• 8/21: Characterization of transistor

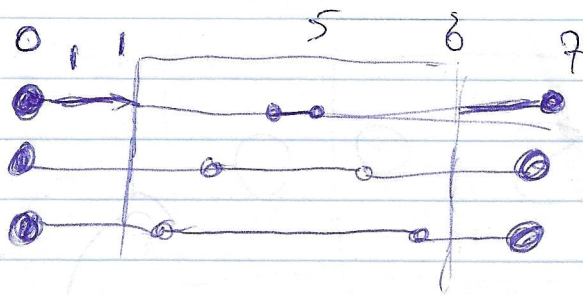
2.5



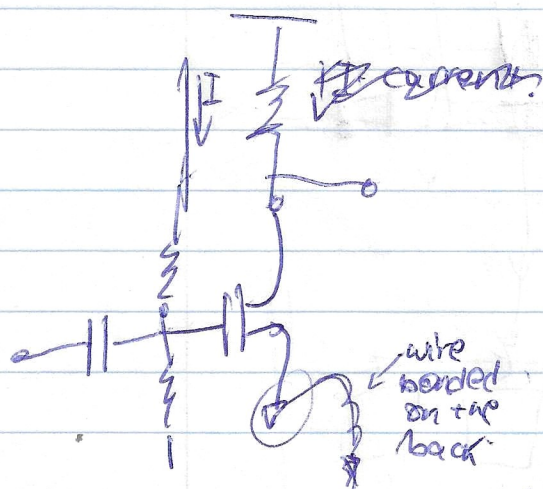
flipping



- common source amplifier.
- common gate drain amplifier.
- characterization of transistor



common source amplifier



$$R = \frac{V}{I}$$

higher voltage test

$$I = \frac{V_{ov}}{R \downarrow}$$

max current

common gate drain amplifier

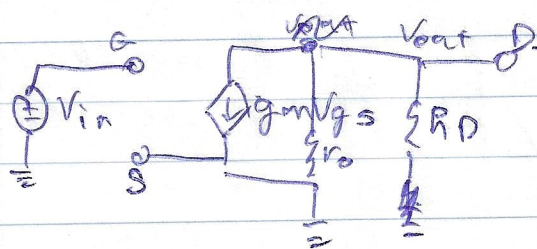
or source follower

or characterization of transistor

→ amplifier.

• CS → used as an amplifier basically:

→ high amplification ✓



$$A_v = \frac{V_{out}}{V_{in}} \Rightarrow V_{out} = 0 - i \cdot R_D$$

Voltage drop

$$i = g_m V_{gs}$$

$$V_{gs} = V_g - V_s = 0 - V_{in} = -V_{in}$$

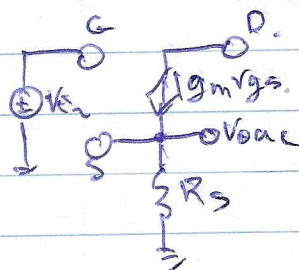
$$i = -g_m V_{in}$$

$$V_{out} = -g_m V_{in} R_D$$

$$V_{out} = -g_m R_D V_{in}$$

• CD → buffer

$$A_v = 1$$



$$V_{out} = g_m V_{gs} R_S = 0$$

$$V_{gs} = V_g - V_s = V_{in} - V_{out}$$

$$V_{out} = g_m (V_{in} - V_{out}) R_S$$

$$V_{out} = g_m V_{in} R_S - g_m V_{out} R_S$$

$$V_{out} (1 + g_m R_S) = g_m R_S V_{in}$$

$$\frac{V_{out}}{V_{in}} = \frac{R_S g_m}{1 + g_m R_S}$$

if $g_m R_S \rightarrow \infty$

$$\rightarrow A_v = 1$$

Signal helper

no amplification

since no low input

high input impedance

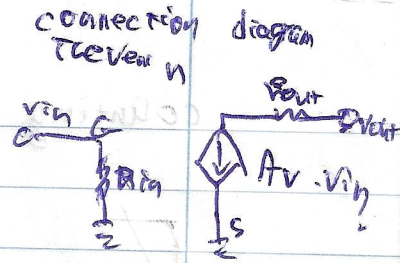
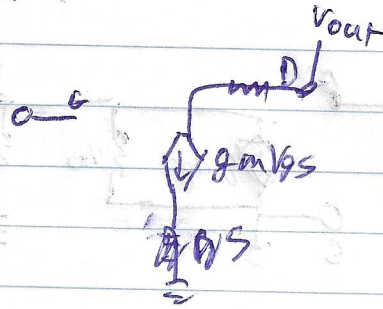
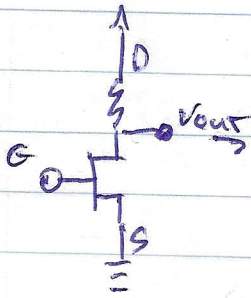
low output impedance

used also for impedance matching.

• temp sensor signal → microphone ADC → ADC input load (weaken)

sensor signal → use CDA to protect signal.

Common source amplifying



$$R_{in} = \frac{V_{test}}{i_{test}}$$

set $v_{in} = V_{test}$

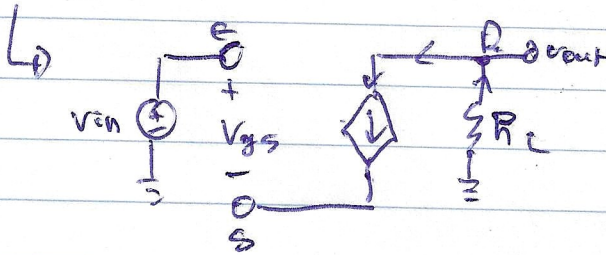
so $i = 0$ since

a small signal $V_{in} \rightarrow$ open

$$R_{in} = \frac{V_{test}}{0} = \infty$$

$$A_v = \frac{v_{out}}{v_{in}}$$

Useful quantity



$$v_{out} = 0 - i \cdot R_L = -i R_L$$

$$i = g_m v_{gs}$$

$$so: v_{out} = -g_m v_{gs} \cdot R_L$$

$$R_{out} = \frac{V_{test}}{i_{test}} \Big|_{v_{in}=0}$$

$v_{in}=0 \rightarrow$ short since $R=0$. $\frac{v_{in}}{i_{in}} = 0$
so $v_{gs} = 0$

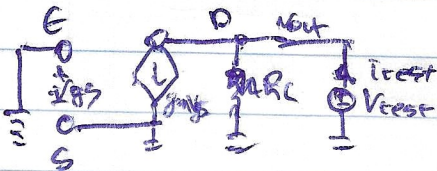
$$v_{gs} = V_g - V_s = 0$$

$$= V_g - 0$$

$$v_{gs} = V_g = v_{in}$$

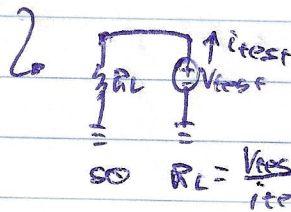
$$so v_{out} = -g_m v_{in} \cdot R_L$$

$$A_v = \frac{v_{out}}{v_{in}} = -g_m R_L$$



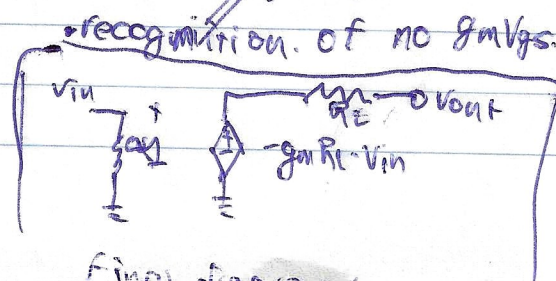
$$v_{gs} = 0$$

$$since i = g_m v_{gs} = 0$$



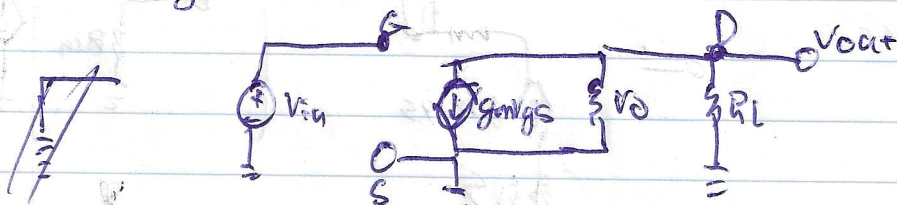
$$so R_L = \frac{V_{test}}{i_{test}} = R_{out}$$

so we can cut out that part of circuit easier to analyze.



Final diagram

counting r_o .



$$A_v = -g_m R_L$$

$$R_{out} = R_L$$

$$R_{in} = \infty$$

before r_o .

→ now
$$A_v = -g_m (R_L // r_o)$$

$$R_{out} = R_L // r_o$$

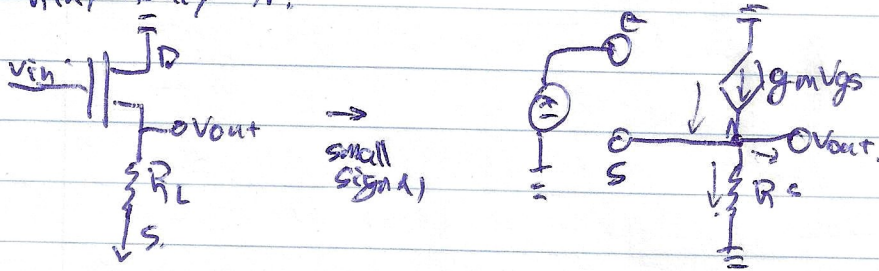
• the reason Small Signal model there is no connection between $G-S$. → r_s because of the structure of the transistor.

• Small signal model of V_{DD} or constant voltage not varied with time is just small signal (gnd).

• that's why.

Common drain amplifier

R_{in} , R_{out} , A_v .



• $R_{in} = \frac{v_{in}}{i_{in}} = \frac{v_{in}}{0} = \infty \rightarrow$ like in CS.

• $v_{out} = g_m v_{gs} R_s = 0$.

• $v_{gs} = v_g - v_s = v_{in} - v_{out}$.

• $v_{out} = g_m (v_{in} - v_{out}) R_s \rightarrow A_v = \frac{v_{out}}{v_{in}} = \frac{g_m R_s}{1 + g_m R_s}$

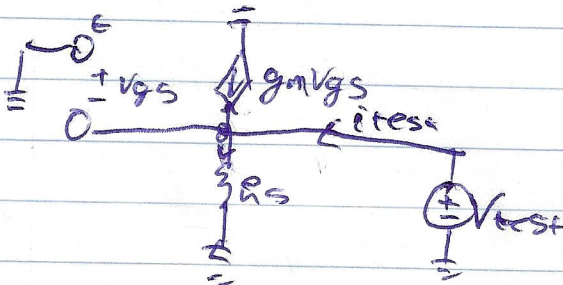
$\hookrightarrow g_m R_s \rightarrow \infty \rightarrow A_v = +1$

$\rightarrow$ good buffer since gain 1

$\rightarrow$ why not just short

$\rightarrow$ because of in/out impedances

• $R_{out} \rightarrow$ test voltage at output.



KCL: $i_{test} = \frac{V_{test}}{R_s} - g_m v_{gs}$

• $v_{gs} = v_g - v_s = 0 - v_s = 0 - V_{test}$

$i_{test} = \frac{V_{test}}{R_s} - g_m (-V_{test})$

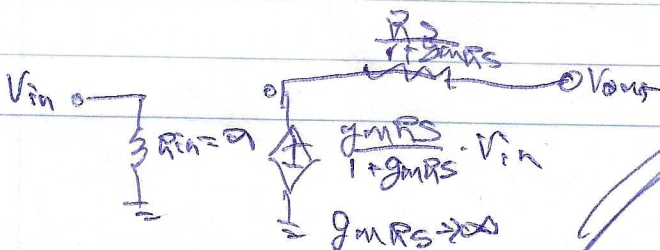
$R_{out} = \frac{V_{test}}{i_{test}} = \frac{1}{\frac{1}{R_s} + g_m}$

R_{out} as $g_m R_s \rightarrow \infty$
if we want it to
operate as buffer.

$\Rightarrow \frac{1}{\frac{1}{R_s} + g_m} \cdot \frac{R_s}{R_s} = \frac{R_s}{1 + g_m R_s} = R_{out}$

$R_{out} = \frac{1}{g_m} \rightarrow$ small value

so high input resistance, $\approx$ gain close to 1 and low output resistance



capac signal capacitance.
~~lateral capacitance~~

C3: ~~lateral~~ $C' = \frac{1}{20V}$ ^{velocity} $C3 = C' \cdot L \rightarrow C3 = 0.138 \text{ pF}$

C1: through ~~lateral~~ vertical capacitance under strip (oxide).

$$C1 = \epsilon_0 \epsilon_r \frac{(W+2S)L}{A} \quad \epsilon_0 = 8.854 \cdot 10^{-12} \quad \dots \rightarrow C1 = 8.92 \text{ fF}$$

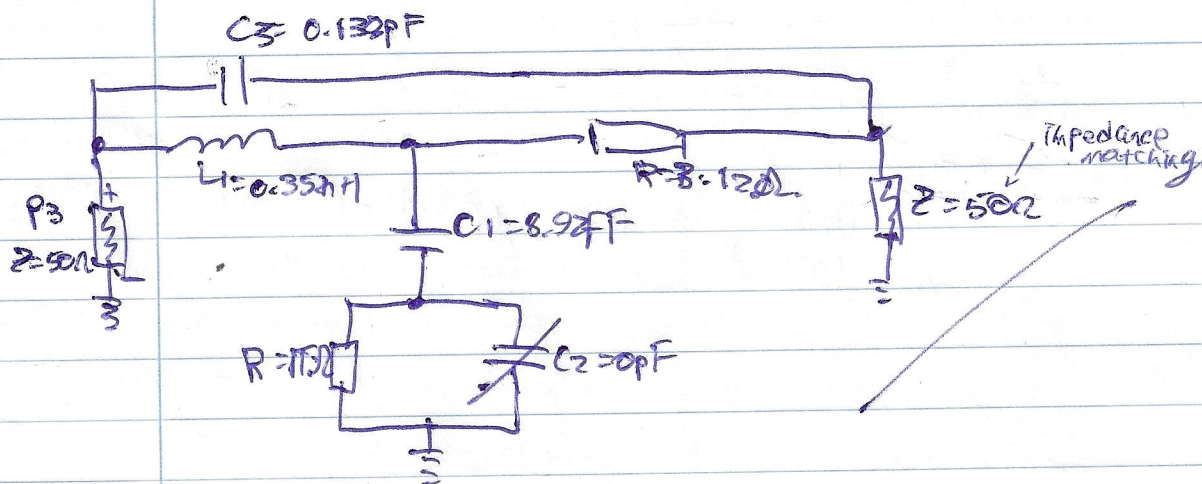
R2: glass is a pretty good substrate compared to Si in such case leakage (modeled as resistance) through glass.

$$R2 = \rho_{\text{glass}} \cdot \frac{H}{(W+2S)L} \quad \left[\begin{array}{l} \text{(borosilicate glass)} \\ \text{commonly used} \\ \text{for interposer} \end{array} \right]$$

glass $\rho = 10^8 - 10^9 \Omega \cdot \text{cm}$.

$\hookrightarrow$ effectively "open" $\rightarrow$ so choosing $10^9 \Omega$.

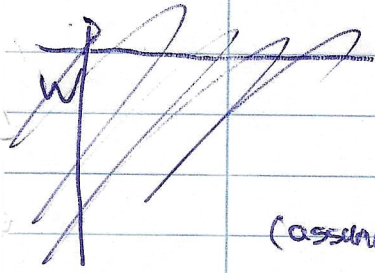
C2: slow wave/space charge $\rightarrow$ irrelevant to glass so. ~~CPF~~.



Goal → Predict Performance of RF die ~~over~~ over a lumped model.
 ↳ extract var sep values from VNA measurements compare →

• Ref - Microwave engineering. → Yeh et

↳ Known Values → reference: Woulter BBEP + CPWG Klayout Design.



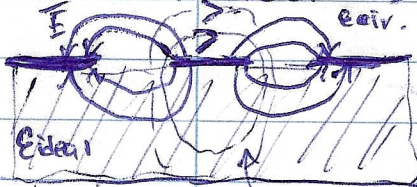
(assume) glass → ϵ_r
 (thickness)

W	80 μm
S	25 μm
L	1mm
H	1mm
	7.75
ϵ_{Au}	100nm
t_{Ti}	50nm
ρ_{Au}	$3.64 \cdot 10^{-8} \Omega\text{m}$
ρ_{Ti}	$3.99 \cdot 10^{-8} \Omega\text{m}$

Parasitics

- R_1 : conductor (Au/Ti) series resistance of 1mm trace.
- L_1 : External + internal inductance of the 1mm CPWG.
- C_3 : lateral (signal ↔ coplanar grounds) capacitance - main CPWG shunt C.
- C_1 : vertical capacitance through glass under strip. ("oxide" term)
- R_2 : leakage through glass. substrate.
- C_2 : slow-wave / space charge capacitance in lossy Si

Parasitical model → simulation (something coplanar line calculator doesn't include.
 ↳ not present in glass //



R_1 DC resistance for each layer.

$$R_i = \rho_i \frac{L}{W t_i} \rightarrow \text{as one on top of other } Au/Ti$$

$$\rightarrow \text{Parallel} \rightarrow R_1 = \left(\frac{1}{\rho_{Au}} + \frac{1}{\rho_{Ti}} \right)^{-1}$$

$$R_1 = 3.12 \Omega //$$

L_1 external + internal inductance of the CPWG:

• external: magnetic field surrounding the current path. / air/glass region.
 (W, S, H) geometry and permeability $\mu_0(\mu_0)$ impact.

• internal: magnetic field → penetrates in metal (finite conductivity of metal).

→ it causes additional inductances at (lower frequency).

↳ (Vanishes as skin effect → pushes it to surface.

↳ not very relevant

• modulus k : $k = \frac{W}{W+2S} = \frac{80\text{nm}}{80\text{nm}+2(25\text{nm})} = 0.615$. $k' = \sqrt{1-k^2} = 0.788$.

• elliptical integral: $K(k) = \int_0^{\pi/2} \frac{d\theta}{\sqrt{1-k^2 \sin^2 \theta}}$ → $K(k) = 1.659$; $K(k') = 1.841 //$

• quasi-static Z-impedance: $Z_0 = \frac{300 \sqrt{k(k')}}{\sqrt{\epsilon_{eff}} K(k)}$ (low frequency impedance)
 (behaves as resistive next to low capacitive effects).

• Phase velocity: $v = \frac{c}{\sqrt{\epsilon_{eff}}} \rightarrow \epsilon_{eff} = \frac{\epsilon_r + 1}{2} = \frac{7.75 + 1}{2} = 4.375$.

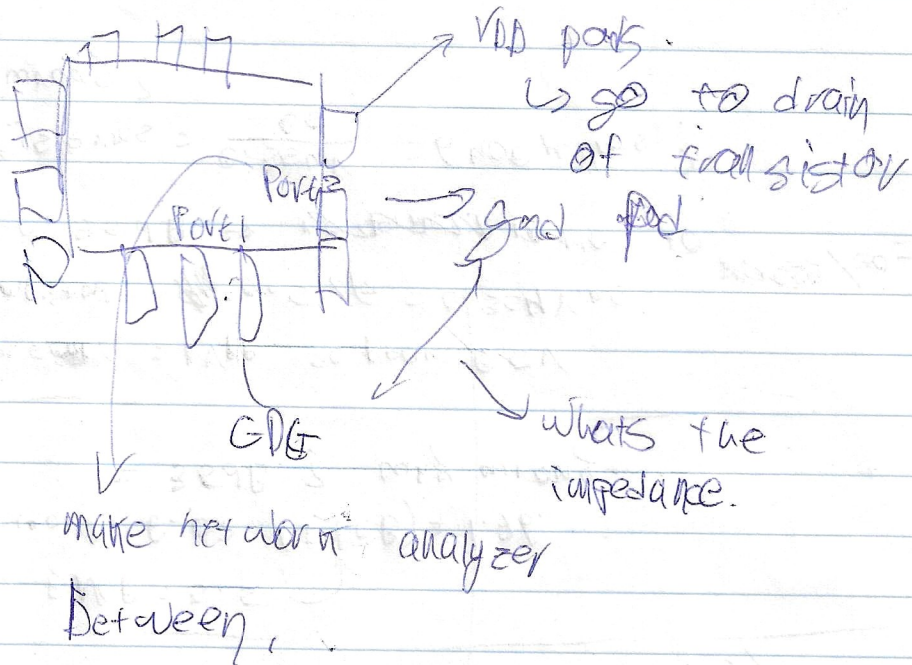
• per unit length inductance: $L' = \frac{Z_0}{v}$

• lumped value: $L = L' \cdot l$

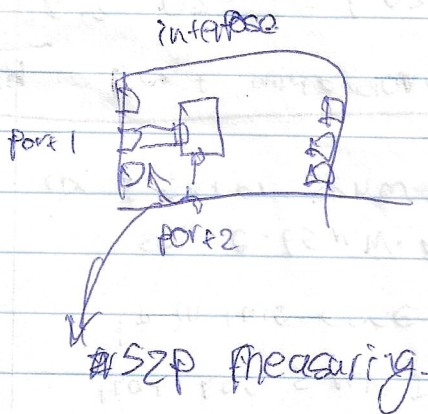
$$L = 1\text{mm}; L' = 3.52 \cdot 10^{-7} \text{H/m}$$

$$\rightarrow L = 0.352 \text{nH} //$$

$Z_c = Z_B + Z_A \rightarrow$ different lengths.



Voskan will provide
a better drawing of details



1. take out pads.
on interposer.
2. separate impact.
of interposer and chip.
 $\rightarrow$ de-embedding.

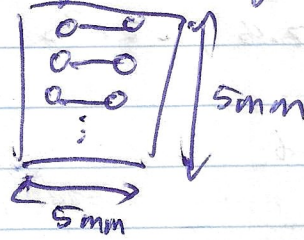
* Separate bump, chip; have interposer.

• with bump

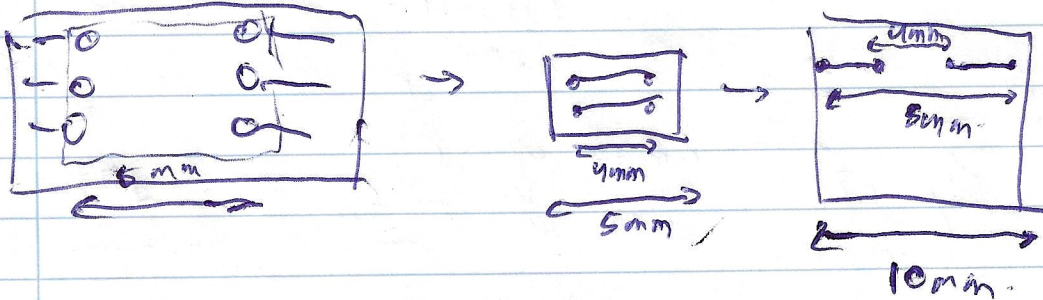
• without bump

* Offset due to bump.

1. - yield testing interposer design sizing change
min glass size 5mm



→ in place the interposer for yield changes too



2. iteration of design for coplanar where
gnd pads ~~and~~ are connected to gnd plane
→ iteration used as final die connect interposer

3. moving pad. short structure

4. design circuit diagram for amplifier die
including smd characterization.
→ include codes.
3 circuits.

Biasing of a common source amplifier.

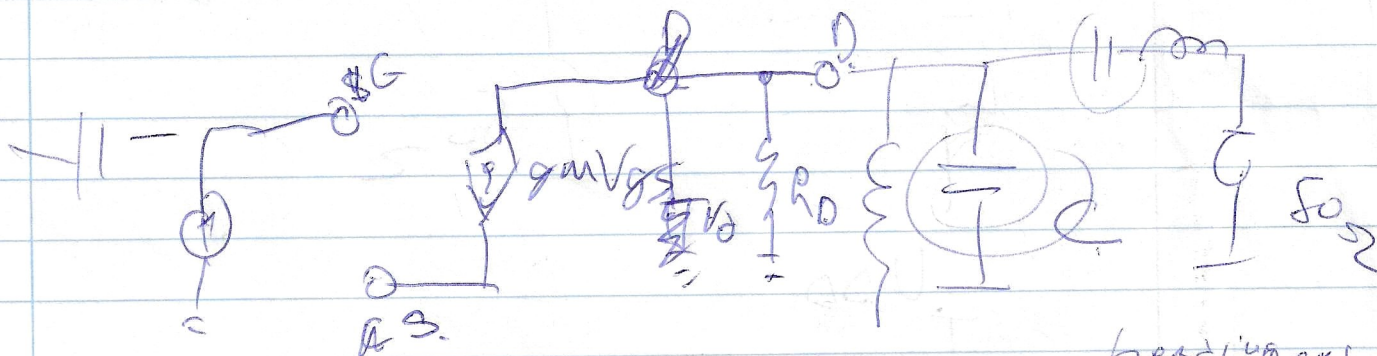
BS2

R_S : $V_S = 2.3V$ $\rightarrow R = \frac{V_S}{I_{D0.05}} \Rightarrow 56\Omega$ (0603)
 $I_D = 50mA$

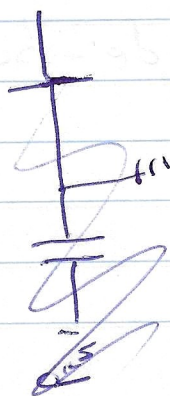
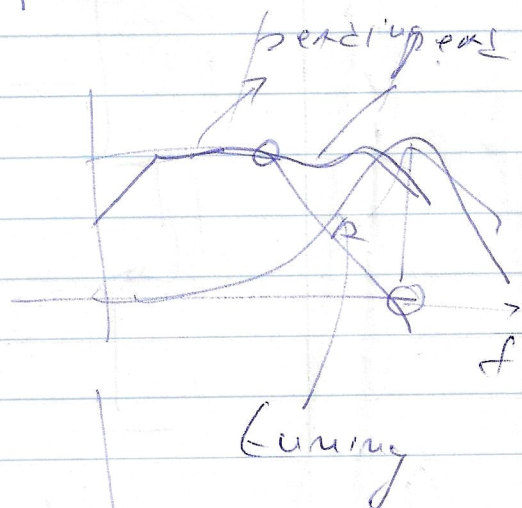
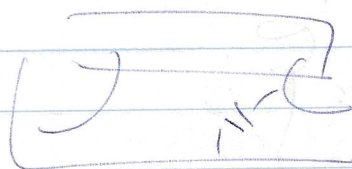
R_D : $V_{DS} = 6V$ $R = \frac{V_{DD} - V_D}{I} = \frac{12 - 2.3}{0.05} = 64\Omega$ (0603)
 $\downarrow$
 mid rail

R_2 gate return $\rightarrow 0V$ (open) $\approx$ large $\geq 10 \cdot R_S$ $1M\Omega$

R_1 weak pull up to set $V_G = 0.1V$ $V_G = V_{DD} \frac{R_2}{R_1 + R_2}$ $R_1 = 10M\Omega$ $V_G = 0.12V$ $10M\Omega$



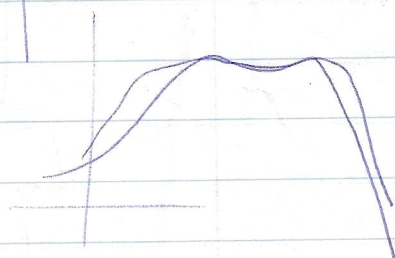
• Inductive peaking.
 • tuning



DC

AC

10 15 10



✓ tuning by shunt inductor
inductive peaking for BW extension

